Bullet Hole Detection on ISSF 10 m Air Rifle Targets

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Assignment 2, Computer Vision (2025/26)

Motivation

Scoring targets is an essential task in **sport shooting**, and a variety of systems have been developed over the years to replace the time-consuming manual scoring. Common technologies include triangulation of acoustic signals, laser-beam interruption and impact sensing. These technologies represent high-accuracy but high-cost solutions, and are well-suited for professional events. Many amateur sport-shooting clubs are interested in **low-cost, real-time scoring systems** that don't require the same level of precision. **Computer-vision-based systems** offer a low-cost alternative capable of detecting and scoring paper targets.

Paper targets may have isolated or overlapping bullet holes. The **overlapping bullet holes** and the **varying lighting conditions** are the main challenges for a computer vision based scoring system.

Dataset

Setup

A camera will capture images of paper targets in the sport shooting club. The camera will be placed on the left side of the shooting line (line between shooter and paper target). Images will be undistorted and warped into plane coordinates. The resolution will be manipulated to result in images of constant shape (1024, 1024).

Figure 1: 10m-air-rifle paper target with isolated and overlapping bullet holes. The image is illustrative.

Data structure

The dataset is divided into train (200), validation (25) and test data (25).

```
dataset/
 images/
    train/
       target_0001.png
       target_0002.png
       ...
    val/
       target_0201.png
       ...
    test/
        target_0226.png
        ...

 annotations.json
 metadata.json
```

The annotations.json file stores for all images in the dataset a list of objects, each described by a class (bullet hole, target black circle) and a bounding box (x, y, w, h).


```
{
  "classes": [
    {"id": 0, "name": "bullet_hole"},
    {"id": 1, "name": "target_black_circle"}
  ],

  "annotations": [
    {
      "image": "target_0001.png",
      "objects": [
        {"class_id": 0, "bbox": [512.4, 487.9, 34.8, 34.8]},
        {"class_id": 0, "bbox": [498.0, 512.5, 35.2, 35.2]},
        {"class_id": 1, "bbox": [512.0, 512.0, 610.0, 610.0]}
      ]
    },
    {
      "image": "target_0002.png",
      "objects": [
        {"class_id": 0, "bbox": [534.2, 480.1, 33.5, 33.5]},
        {"class_id": 1, "bbox": [512.0, 512.0, 610.0, 610.0]}
      ]
    }
  ]
}
```


The metadata.json file has information like image size, pixel resolution, and relevant paths to access the images.


```
{
  "image_size_px": [1024, 1024],
  "pixels_per_mm": 10.0,
  "train_path": "dataset/images/train/",
```

```
  "val_path":   "dataset/images/val/",
  "test_path":  "dataset/images/test/"
}
```

Challenge

1. Propose a custom AI model that performs bullet center and target center detection from the images.
2. Train and test the AI model without and with data augmentation.
3. Visualize the results in a virtual target constructed from the known 10m-air-rifle-target dimensions and from the relative positions of the bullet centers with respect to the target center.
4. Use the following metrics to assess the quality of your implementation:
    - RMSE of training and testing;
    - Confusion matrix;
    - Model complexity (number of parameters).
5. Compare the results of your AI model with at least one existing model (e.g., MobileNet, VGG, EfficientNet, or ResNet).
6. Discuss the obtained results.